

Akshat Gupta

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EDUCATION

KIET GROUP of Institutions - AKTU

Bachelors in Computer Science with specialization in AI (GPA: 7.8)

Ghaziabad, Uttar Pradesh

Nov. 2022 – May 2026

Maa Bharti International - CBSE

Intermediate (Class 12th) - 85%

Kota, Rajasthan

May. – Mar 2022

EXPERIENCE

Mark AI | Software Developer | Full Time | Platform

Jun 2026 – Present

- Built an AI-native **DOOH CMS** using **FastAPI** with intelligent campaign management, scheduling, payment gateway integration, and a **Kotlin** Android player for remote content delivery across **1Lakh+ digital displays**.
- Built an **AI-powered booking agent** that leverages **ML-based recommendation models** to suggest **related** and **competitor screens** while converting **natural-language** requests into executable booking scripts, improving campaign targeting, screen discovery, and the overall booking experience.
- Designed and developed an interactive **3D website** using **Blender** and **Three.js/WebGL** for immersive screen visualization.
- Integrated **AR visualization** using **MindAR** and **3D human models**, enabling clients to preview advertisements in real-world environments through marker-based scanning before campaign deployment.

VibeMonitor | Software Developer | Full Time | Platform

Sep 2025 – Jun 2026

- Built **Beryl**, an **AI-driven end-to-end test-automation** product that crawls any URL (including **authenticated** pages), generates **natural-language** test cases, converts them into **Playwright** scripts, and runs them in a **headless browser** to report pass/fail results and a site-reliability score.
- Optimized **CI/CD pipelines** with multi-stage workflows, build caching, and artifact reuse, reducing build time by **5%**. Automated **RDS snapshot** management for backups and environment migrations, and built an automated **RCA engine** that traces incidents through commit history to accelerate root-cause analysis and debugging.
- Built a scalable **log ingestion platform** with SDKs in **Java, Python, and JavaScript** to stream high-volume logs into **ClickHouse** using efficient batching and asynchronous pipelines.
- Engineered a centralized **observability platform** integrating **AWS CloudWatch, New Relic, Datadog, Grafana**, and custom **Prometheus** metrics, applying **RED** and **USE** methodologies for unified monitoring, alerting, and proactive service health tracking across multiple microservices.

Sakhi Women | Full Stack Developer | Freelance | Platform

Jan 2026 – Feb 2026

- Built ride-booking platform with **RBAC, OTP-based passwordless auth**, and time-based **dynamic pricing** (1.5x night multiplier), supporting **1500-2000 concurrent users** through **B-tree indexing** on ride-matching and booking queries, ensuring consistent performance under load.
- Integrated **Twilio WhatsApp API** for multi-stage notification workflows (OTP, assignments, driver alerts) and implemented **cron-based recurring scheduling** for subscription rides with automated driver assignment logic.

iTax Easy | Full Stack and Server Developer | Internship | Platform

Feb 2025 – Sep 2025

- Built high-performance web and mobile applications using **PostgreSQL, NextJS, Prisma, React Native**, and **hPanel**, decreasing load times and increasing user retention.
- Managed production infrastructure on **Hostinger HPanel**, handling deployments, server configuration, SSL setup, and database operations. Integrated **Income Tax Sandbox APIs** to automate taxpayer verification and compliance workflows.

PrepSaarthi | Co-Founder & Technical Lead (Mentor-based Startup)

Nov. 2024 – Jul 2025

- Co-founded an edtech platform focused on mentorship, career guidance, and student upskilling.
- Developed 5+ production features using **React.js** and **MongoDB**, implementing database indexing and secure API integrations. Managed production deployments, server infrastructure, and application monitoring on **Hostinger HPanel**.

PROJECTS

Loupe — Screen Recorder & Video Editor | Electron, Swift, .NET 8, WebCodecs, Whisper, ONNX Live | Aug 2026 – Present

- Building a **cross-platform** screen recorder for **macOS and Windows** that zooms to **4x** toward the cursor, with the native capture layer written twice behind one protocol in **Swift** and **.NET 8**.
- Built the **non-linear editor without FFmpeg**, encoding and muxing through **WebCodecs**, with **0.25x–8x WSOLA** speed ramps that hold natural pitch and on-device **Whisper** captions.

Cosmic-Trek (Orrery-App) | ThreeJs, pynb, WebGL, D3Js, HTML, CSS, Javascript, Redux

Dec 2025 – Mar 2026

- Constructed interactive **3D solar system simulator** featuring 8+ celestial bodies and **50+ exoplanets**, achieving **60%** reduction in rendering time.
- Programmed **physics-based satellite launch simulator** with 5+ orbital patterns, decreasing computation load by **30%**.
- Created realistic collision and gravitational simulations with 10+ custom impact effects, resulting in **75%** increase in user engagement.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, C++, Kotlin

Frontend: ReactJS, NextJS, Angular, Redux, Three.js, Electron, Tailwind CSS, Vite

Backend, Data & AI: FastAPI, NodeJS, ExpressJS, Prisma, REST, WebSocket, OAuth 2.0, JWT, PostgreSQL, MongoDB, ClickHouse, Redis, Kafka, RAG, LangGraph, OpenAI API, Amazon Bedrock, Whisper, ONNX Runtime

Cloud, DevOps & Observability: AWS(EKS, ECS, EC2, Fargate, lambda, EBS, EFS, Cloudwatch), GCP, Docker, CI/CD, Vercel, Firebase, GitHub Actions, Git, Datadog, New Relic, Grafana, Prometheus, OpenTelemetry, Loki, Signoz, Playwright